

Putnam Training (Sep 22)

Problem 1 (2021/B2). Determine the maximum value of the sum

$$S = \sum_{n=1}^{\infty} \frac{n}{2^n} (a_1 a_2 \dots a_n)^{\frac{1}{n}}$$

over all sequences $a_1, a_2, a_3, \dots$ of nonnegative real numbers satisfying

$$\sum_{k=1}^{\infty} a_k = 1.$$

Problem 2 (2005/B2). Find all positive integers $n, k_1, \dots, k_n$ such that $k_1 + \dots + k_n = 5n - 4$ and

$$\frac{1}{k_1} + \dots + \frac{1}{k_n} = 1.$$

Problem 3 (2002/B3). Show that for all integers $n > 1$,

$$\frac{1}{2ne} < \frac{1}{e} - \left(1 - \frac{1}{n}\right)^n < \frac{1}{ne}.$$

Problem 4 (2013/B4). For any continuous real-valued function f defined on the interval $[0, 1]$, let

$$\mu(f) = \int_0^1 f(x) dx, \quad \text{Var}(f) = \int_0^1 (f(x) - \mu(f))^2 dx,$$

$$M(f) = \max_{0 \leq x \leq 1} |f(x)|.$$

Show that if f and g are continuous real-valued functions defined on the interval $[0, 1]$, then

$$\text{Var}(fg) \leq 2\text{Var}(f)M(g)^2 + 2\text{Var}(g)M(f)^2.$$

Problem 5 (2014/A4). Suppose X is a random variable that takes on only non-negative integer values, with $E[X] = 1$, $E[X^2] = 2$ and $E[X^3] = 5$. (Here $E[Y]$ denotes the expectation of the random variable Y .) Determine the smallest possible value of the probability of the event $X = 0$.

Problem 6 (2022/B5). For $0 \leq p \leq 1/2$, let $X_1, X_2, \dots$ be independent random variables such that

$$X_i = \begin{cases} 1 & \text{with probability } p, \\ -1 & \text{with probability } p, \\ 0 & \text{with probability } 1 - 2p, \end{cases}$$

for all $i \geq 1$. Given a positive integer n and integers $b, a_1, \dots, a_n$, let $P(b, a_1, \dots, a_n)$ denote the probability that $a_1 X_1 + \dots + a_n X_n = b$. For which values of p is it the case that

$$P(0, a_1, \dots, a_n) \geq P(b, a_1, \dots, a_n)$$

for all positive integers n and all integers $b, a_1, \dots, a_n$?